Importing Library and Importing the sample file

In [1]:
```
import pandas as pd

emp_data = pd.read_csv("sample_employee_dataset_.csv")
emp_data
```

Out[1]:

	Employee ID	Name	Age	Department	Salary ($)	Years at Company	Performance Score	Attrition
0	E0001	Oscar Walker	35	Finance	83073	17	4.2	Yes
1	E0002	Hannah Patel	43	Finance	61105	19	4.6	No
2	E0003	Nina Scott	26	Human Resources	54998	6	4.2	No
3	E0004	Bob Clark	55	Customer Support	106850	11	4.0	No
4	E0005	Hannah Hall	59	Finance	55421	0	4.5	No
...	...	...	...	...	...	...	...	...
3249	E3250	Nina Kim	46	Marketing	53870	9	4.1	No
3250	E3251	Dana Kim	40	Finance	79577	3	3.3	No
3251	E3252	Nina Scott	36	Customer Support	84511	7	4.0	No
3252	E3253	Oscar Allen	49	Customer Support	72456	13	2.8	No
3253	E3254	Oscar Lopez	55	Customer Support	81901	18	3.7	No

3254 rows × 8 columns

Checking the duplicate the values and removing the unnecessary columns

In [2]:
```
print(emp_data.duplicated().sum())
```

```
0
```

In [3]:
```
emp_data
```

Out[3]:

	Employee ID	Name	Age	Department	Salary ($)	Years at Company	Performance Score	Attrition
0	E0001	Oscar Walker	35	Finance	83073	17	4.2	Yes
1	E0002	Hannah Patel	43	Finance	61105	19	4.6	No
2	E0003	Nina Scott	26	Human Resources	54998	6	4.2	No
3	E0004	Bob Clark	55	Customer Support	106850	11	4.0	No
4	E0005	Hannah Hall	59	Finance	55421	0	4.5	No
...	...	...	...	...	...	...	...	...
3249	E3250	Nina Kim	46	Marketing	53870	9	4.1	No
3250	E3251	Dana Kim	40	Finance	79577	3	3.3	No
3251	E3252	Nina Scott	36	Customer Support	84511	7	4.0	No
3252	E3253	Oscar Allen	49	Customer Support	72456	13	2.8	No
3253	E3254	Oscar Lopez	55	Customer Support	81901	18	3.7	No

3254 rows × 8 columns

Performing Exploratory Data Analysis (EDA) (calculating Mean,Median,Mode)

In [4]:
```
corr = emp_data.describe()
corr
```

Out[4]:

	Age	Salary ($)	Years at Company	Performance Score
count	3254.000000	3254.000000	3254.000000	3254.000000
mean	41.301168	83139.367548	10.055624	4.001076
std	11.185066	22009.555668	6.031196	0.494528
min	22.000000	45005.000000	0.000000	2.100000
25%	32.000000	64350.000000	5.000000	3.700000
50%	41.000000	83342.000000	10.000000	4.000000
75%	51.000000	102467.000000	15.000000	4.300000
max	60.000000	119996.000000	20.000000	5.000000

This result suggest that the Avarage age of people in the company is 41 years among which 25% are under 32 year,50% are under 41 year and 75% are 51 years,they serve the company in an average of 10 years where as 25% have serve less than 5 years, 50% have serve almost 10 years and 75% have severd for 15 years with the performance score of 4 in which 25% employee have performance score of 3.7, 50% median score of 4 and 75% have les than 4.3.

the maximum age of the employee is 60, Highest salary is almost 120000 and max year of serving in company is 20 years with the performance score of 5

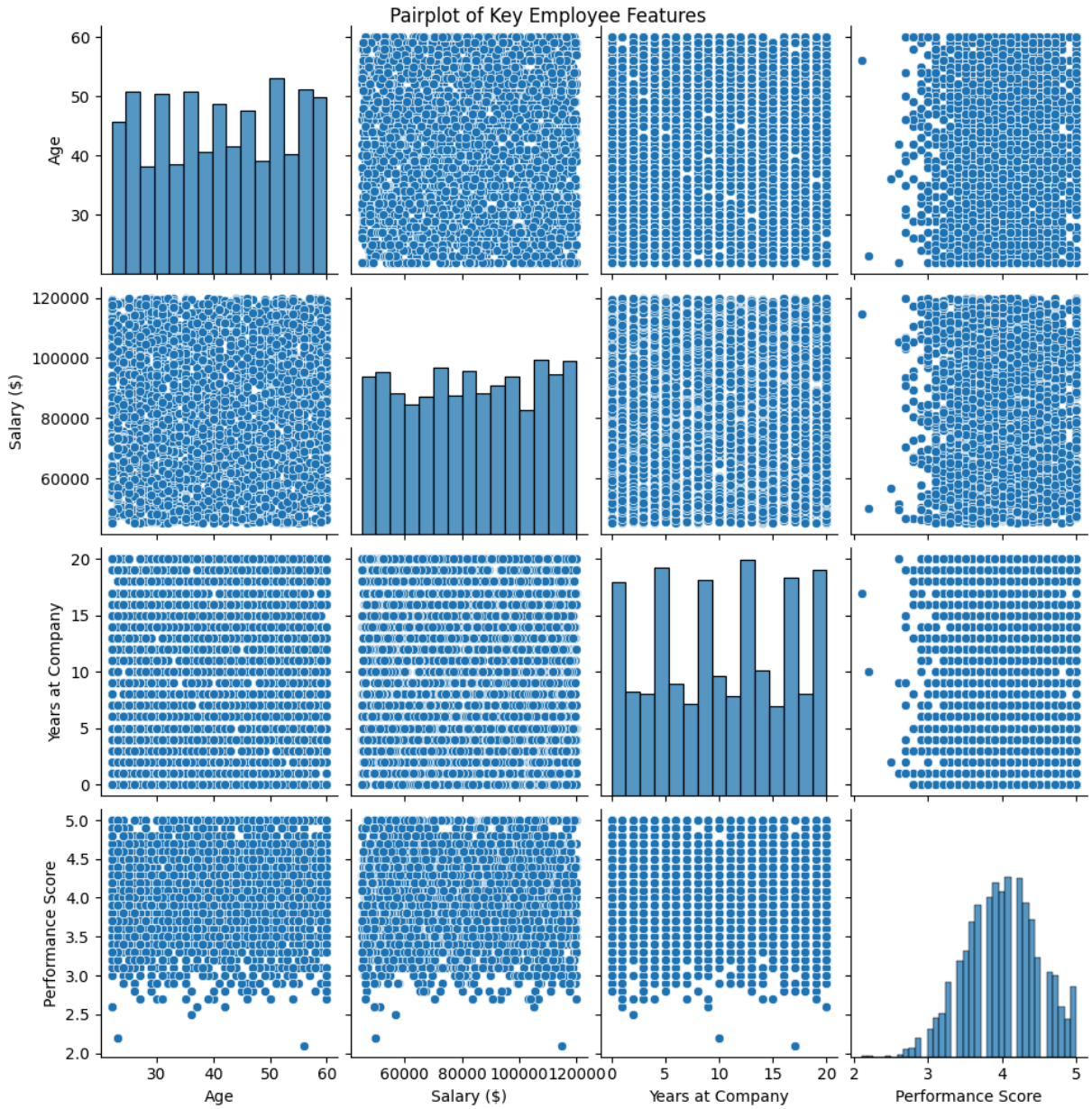
In [5]:
```
corr_emp_data=emp_data[['Age','Salary ($)','Years at Company','Performance Score']]
corr_emp_data
```

Out[5]:

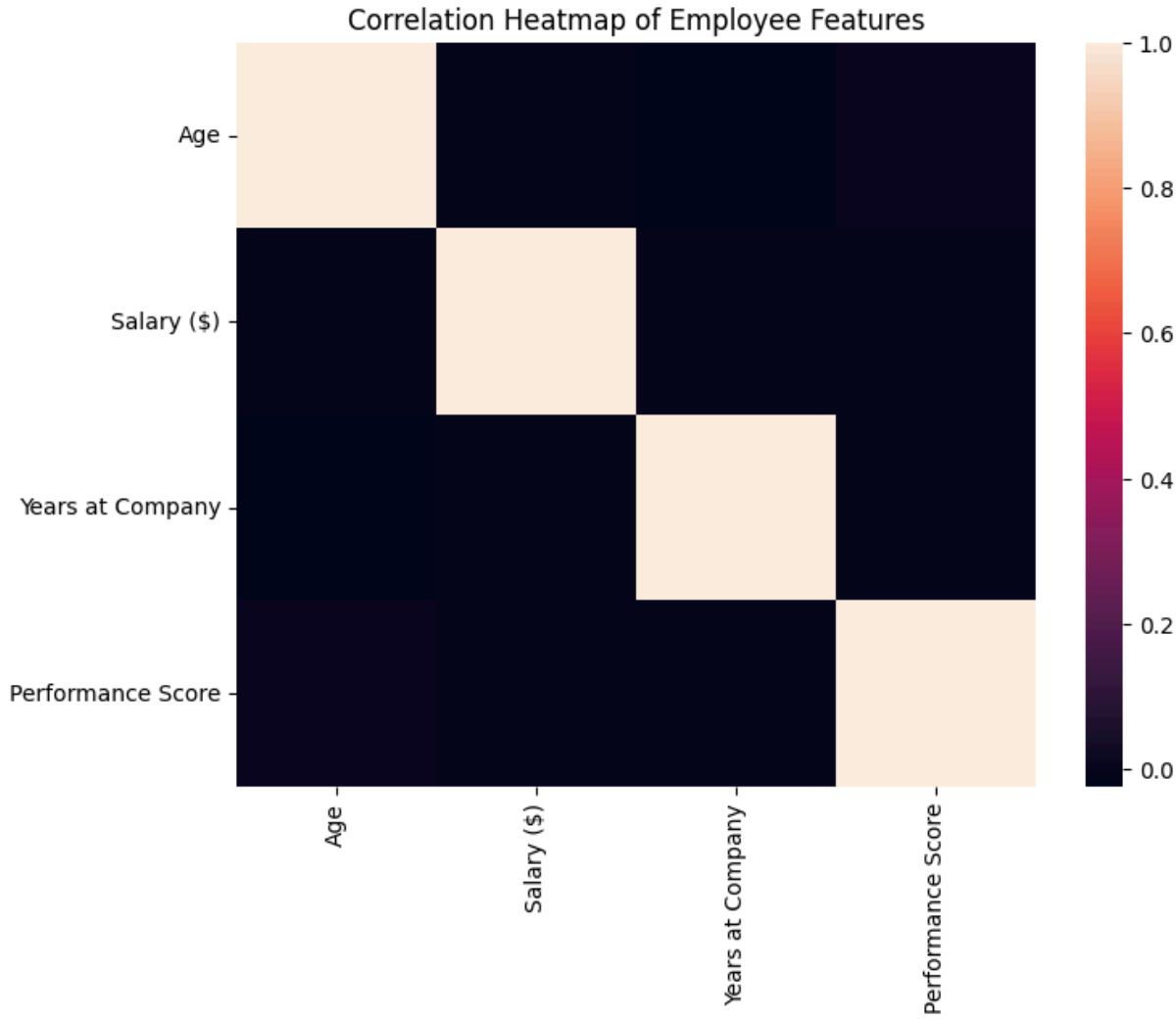
	Age	Salary ($)	Years at Company	Performance Score
Age	1.000000	-0.016092	-0.024099	0.006733
Salary ($)	-0.016092	1.000000	-0.004389	-0.005688
Years at Company	-0.024099	-0.004389	1.000000	-0.008523
Performance Score	0.006733	-0.005688	-0.008523	1.000000

In [6]:
```
import seaborn as sns
import matplotlib.pyplot as plt

sns.pairplot(emp_data[['Age','Salary ($)','Years at Company','Performance Score']])
```

```
plt.suptitle('Pairplot of Key Employee Features', y=1)
plt.show()
```

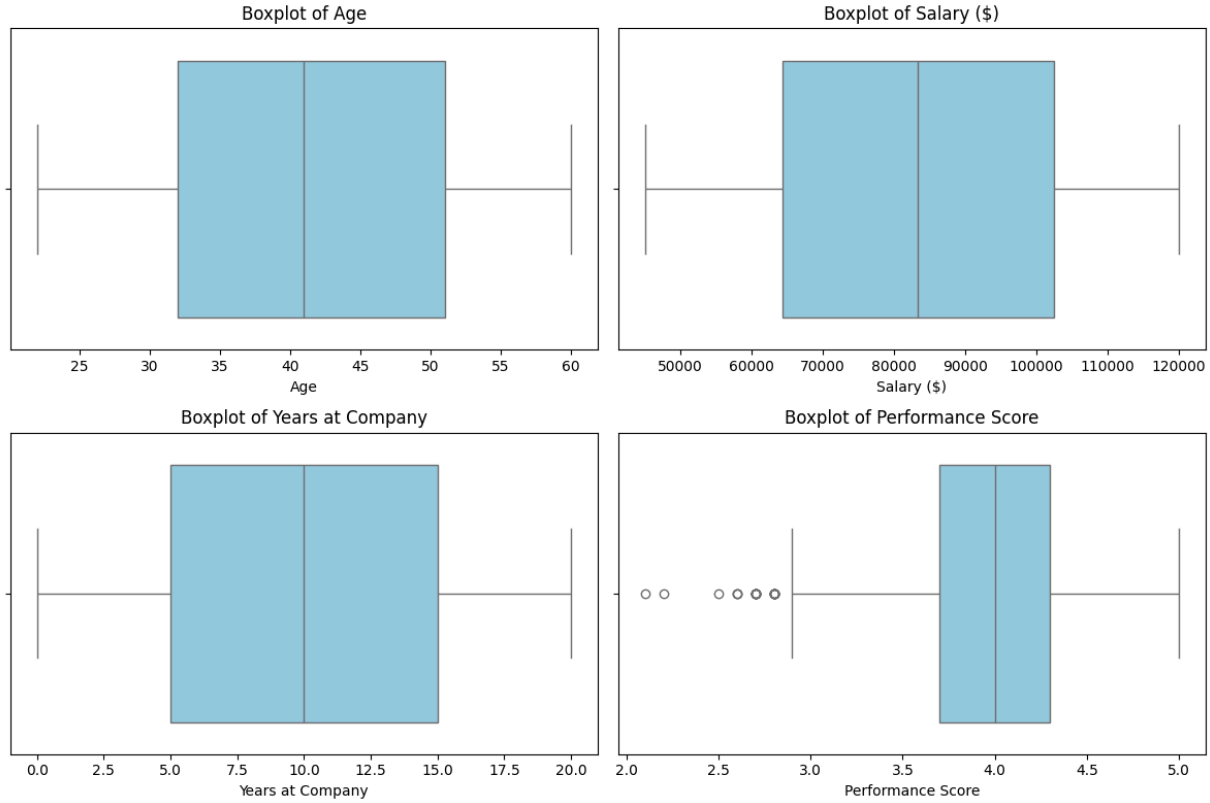


In [7]:
```
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(corr_emp_data)
plt.title('Correlation Heatmap of Employee Features')
plt.show()
```



BoxPlot

In [8]:
```
numeric_cols = ['Age', 'Salary ($)', 'Years at Company', 'Performance Score']

plt.figure(figsize=(12, 8))

for i, col in enumerate(numeric_cols, start=1):
    plt.subplot(2, 2, i)
    sns.boxplot(x=emp_data[col], color='skyblue', whis=1.5)
    plt.title(f'Boxplot of {col}')

plt.tight_layout()
plt.show()
```



Probability of Leaving

In [9]:
```
emp_data['Left'] = emp_data['Attrition'].map({'Yes':1, 'No':0})
emp_data['Perf_bin'] = pd.cut(emp_data['Performance Score'], bins=[0,3,4,5], labels
prob_df = (
    emp_data.groupby(['Department','Perf_bin'])['Left']
    .agg(['mean','count'])
    .rename(columns={'mean':'Prob_left','count':'N'})
    .reset_index()
)
print(prob_df)
```

```
          Department Perf_bin  Prob_left    N
0   Customer Support      Low   0.076923   13
1   Customer Support   Medium   0.214912  228
2   Customer Support     High   0.178082  219
3        Engineering      Low   0.111111    9
4        Engineering   Medium   0.200000  250
5        Engineering     High   0.240741  216
6            Finance      Low   0.066667   15
7            Finance   Medium   0.211009  218
8            Finance     High   0.198980  196
9    Human Resources      Low   0.214286   14
10   Human Resources   Medium   0.241379  232
11   Human Resources     High   0.232759  232
12                IT      Low   0.111111    9
13                IT   Medium   0.169421  242
14                IT     High   0.175115  217
15         Marketing      Low   0.125000   16
16         Marketing   Medium   0.252212  226
17         Marketing     High   0.246445  211
18             Sales      Low   0.157895   19
19             Sales   Medium   0.181034  232
20             Sales     High   0.220833  240
```

```
C:\Users\tbira\AppData\Local\Temp\ipykernel_15744\401102564.py:4: FutureWarning: The
default of observed=False is deprecated and will be changed to True in a future vers
ion of pandas. Pass observed=False to retain current behavior or observed=True to ad
opt the future default and silence this warning.
  emp_data.groupby(['Department','Perf_bin'])['Left']
```

Bayes Therom

In [10]:

```python
p_left = emp_data['Left'].mean()
likelihood = (
    emp_data[emp_data['Left']==1]
    .groupby('Perf_bin')['Left']
    .count() / emp_data['Left'].sum()
)
evidence = emp_data.groupby('Perf_bin')['Left'].count() / len(emp_data)
bayes = (likelihood * p_left) / evidence
print(bayes)
```

```
Perf_bin
Low       0.126316
Medium    0.209459
High      0.213586
Name: Left, dtype: float64
```

```
C:\Users\tbira\AppData\Local\Temp\ipykernel_15744\2461238685.py:4: FutureWarning: Th
e default of observed=False is deprecated and will be changed to True in a future ve
rsion of pandas. Pass observed=False to retain current behavior or observed=True to
adopt the future default and silence this warning.
  .groupby('Perf_bin')['Left']
C:\Users\tbira\AppData\Local\Temp\ipykernel_15744\2461238685.py:7: FutureWarning: Th
e default of observed=False is deprecated and will be changed to True in a future ve
rsion of pandas. Pass observed=False to retain current behavior or observed=True to
adopt the future default and silence this warning.
  evidence = emp_data.groupby('Perf_bin')['Left'].count() / len(emp_data)
```

Scaling Numerical Features

In [11]:
```
from sklearn.preprocessing import StandardScaler
num_features = ['Age', 'Salary ($)', 'Years at Company', 'Performance Score']
scaler = StandardScaler()
emp_data[num_features] = scaler.fit_transform(emp_data[num_features])
```

Labeling and Encoding (Categorical data)

In [12]:
```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
emp_data['Attrition_encoded'] = le.fit_transform(emp_data['Attrition'])
emp_data['Department_encoded'] = le.fit_transform(emp_data['Department'])
emp_data_ml = emp_data.drop(columns=['Attrition', 'Department', 'Perf_bin'])
print(emp_data_ml.head())
```

```
  Employee ID          Name       Age  Salary ($)  Years at Company  \
0       E0001  Oscar Walker -0.563442   -0.003016          1.151586
1       E0002  Hannah Patel  0.151907   -1.001281          1.483247
2       E0003    Nina Scott -1.368210   -1.278794         -0.672544
3       E0004     Bob Clark  1.224931    1.077454          0.156606
4       E0005   Hannah Hall  1.582606   -1.259572         -1.667525

   Performance Score  Left  Attrition_encoded  Department_encoded
0           0.402313     1                  1                   2
1           1.211290     0                  0                   2
2           0.402313     0                  0                   3
3          -0.002175     0                  0                   0
4           1.009046     0                  0                   2
```

Train/Test Dataset

In [13]:
```
from sklearn.model_selection import train_test_split
X = emp_data_ml.drop(columns=['Left', 'Employee ID','Name'])
y = emp_data_ml['Left']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta
```

In [14]:
```
print(emp_data_ml.columns)
```

```
Index(['Employee ID', 'Name', 'Age', 'Salary ($)', 'Years at Company',
       'Performance Score', 'Left', 'Attrition_encoded', 'Department_encoded'],
      dtype='object')
```

In [ ]:

Trainnin Classification Model

In [15]:
```
from sklearn.ensemble import RandomForestClassifier
rf_model = RandomForestClassifier(random_state=42)
rf_model.fit(X_train, y_train)
y_pred = rf_model.predict(X_test)
```

Evaluatin the modle

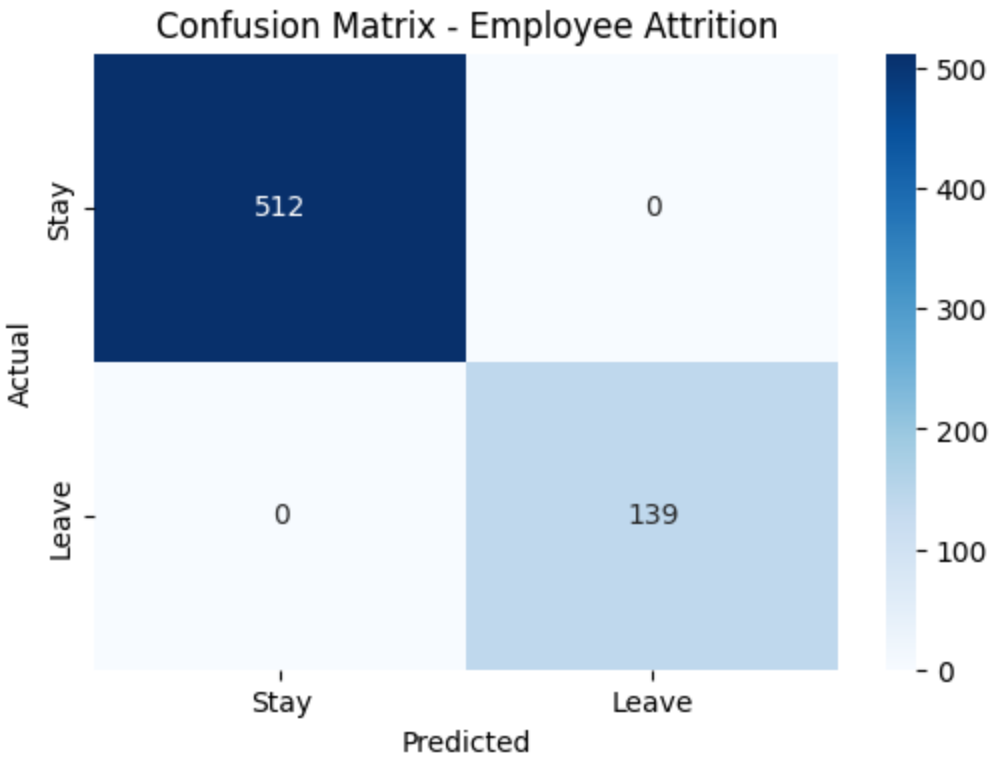
In [16]:
```
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

print("Accuracy:", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall:", recall_score(y_test, y_pred))
print("F1 Score:", f1_score(y_test, y_pred))

print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
Accuracy: 1.0
Precision: 1.0
Recall: 1.0
F1 Score: 1.0

Classification Report:
               precision    recall  f1-score   support

           0       1.00      1.00      1.00       512
           1       1.00      1.00      1.00       139

    accuracy                           1.00       651
   macro avg       1.00      1.00      1.00       651
weighted avg       1.00      1.00      1.00       651
```

visualization

In [17]:
```
import seaborn as sns
import matplotlib.pyplot as plt

cm = confusion_matrix(y_test, y_pred)
plt.figure(figsize=(6, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Stay', 'Leave'], yticklabels=['Stay', 'Leave'])
plt.xlabel('Predicted')
```

```
plt.ylabel('Actual')
plt.title('Confusion Matrix - Employee Attrition')
plt.show()
```



Prepare Data

In [18]:
```
from sklearn.model_selection import train_test_split


y_perf = emp_data_ml['Performance Score']
X_perf = emp_data_ml.drop(columns=['Employee ID','Performance Score','Name'])
X_train_p, X_test_p, y_train_p, y_test_p = train_test_split(X_perf, y_perf, test_si
```

usin linear reression model

In [19]:
```
from sklearn.linear_model import LinearRegression

lr_model = LinearRegression()
lr_model.fit(X_train_p, y_train_p)

y_pred_p = lr_model.predict(X_test_p)
```

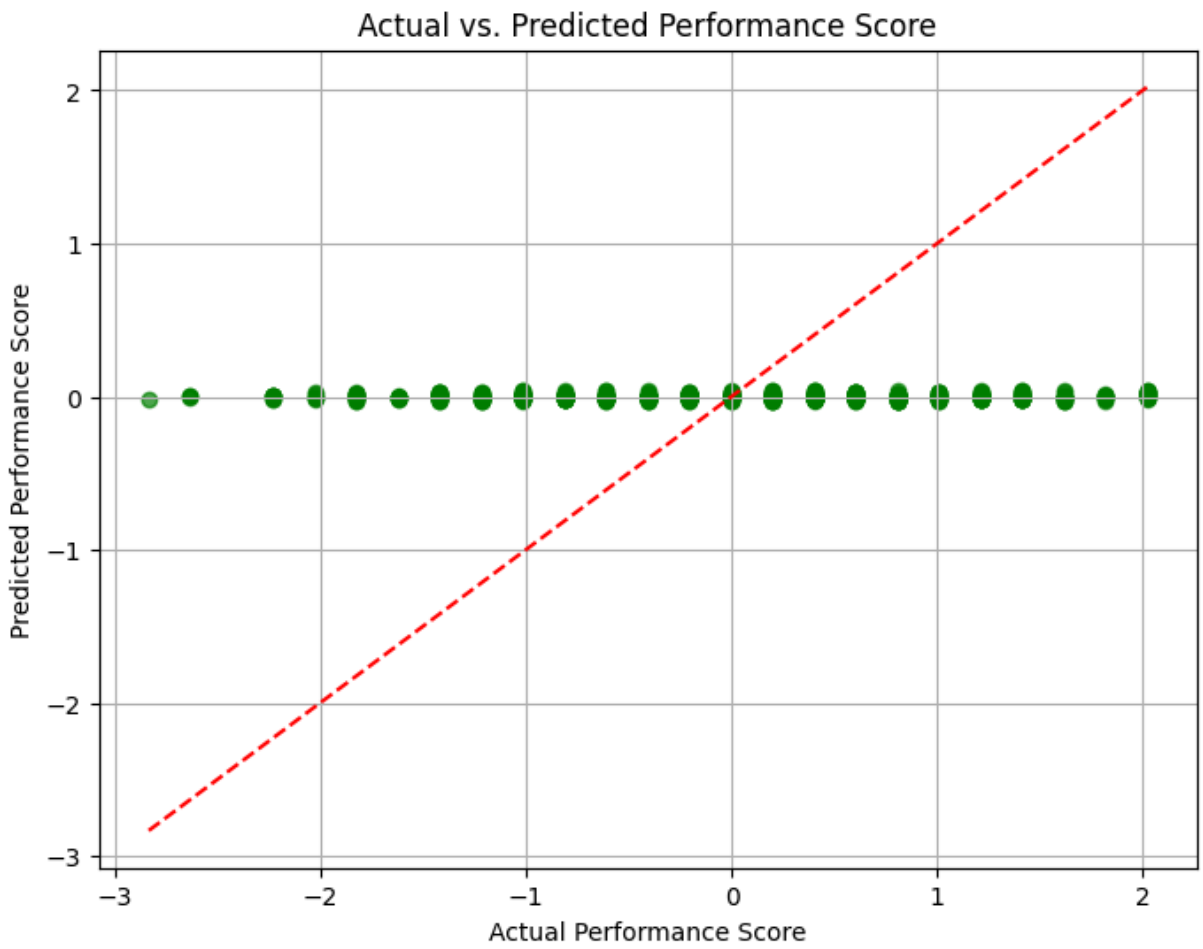
Evaluation

In [20]:
```
from sklearn.metrics import r2_score, mean_squared_error

r2 = r2_score(y_test_p, y_pred_p)
mse = mean_squared_error(y_test_p, y_pred_p)

print("R-squared (R²):", r2)
print("Mean Squared Error (MSE):", mse)
```

```
R-squared (R²): 0.0016524071100597615
Mean Squared Error (MSE): 1.022444774733074
```

visualation

In [21]:
```
plt.figure(figsize=(8, 6))
plt.scatter(y_test_p, y_pred_p, alpha=0.7, color='green')
plt.plot([y_test_p.min(), y_test_p.max()], [y_test_p.min(), y_test_p.max()], 'r--')
plt.xlabel('Actual Performance Score')
plt.ylabel('Predicted Performance Score')
plt.title('Actual vs. Predicted Performance Score')
plt.grid(True)
plt.show()
```



In [22]:
```
from sklearn.linear_model import LinearRegression
```

Feedforward

In [23]:
```
import tensorflow as tf
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
from sklearn.preprocessing import StandardScaler

X = emp_data_ml.drop(columns=['Employee ID','Performance Score','Name'])
y = emp_data_ml['Performance Score']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = Sequential([
    Dense(64, input_dim=X_train.shape[1], activation='relu'),
    Dense(32, activation='relu'),
    Dense(1)
])

model.compile(optimizer='adam', loss='mean_squared_error', metrics=['mae'])

history = model.fit(X_train, y_train, epochs=15, batch_size=32, validation_split=0.
```

```
---------------------------------------------------------------------------
ModuleNotFoundError                       Traceback (most recent call last)
Cell In[23], line 1
----> 1 import tensorflow as tf
      2 from tensorflow.keras.models import Sequential
      3 from tensorflow.keras.layers import Dense

ModuleNotFoundError: No module named 'tensorflow'
```

Evaluate

In [ ]:
```
loss, mae = model.evaluate(X_test, y_test)
print(f"Test MSE: {loss}")
print(f"Test MAE: {mae}")
```

Plot

In [ ]:
```
y_pred_nn = model.predict(X_test).flatten()

plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred_nn, alpha=0.7, color='purple')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--')
plt.xlabel('Actual Performance Score')
```

```
plt.ylabel('Predicted (NN)')
plt.title('Neural Network - Actual vs Predicted Performance')
plt.grid(True)
plt.show()
```